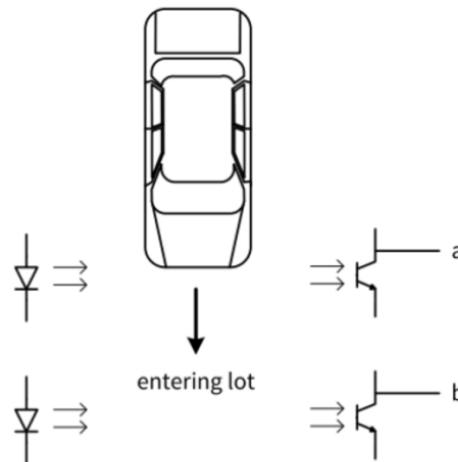


CAD-HW04-exe2

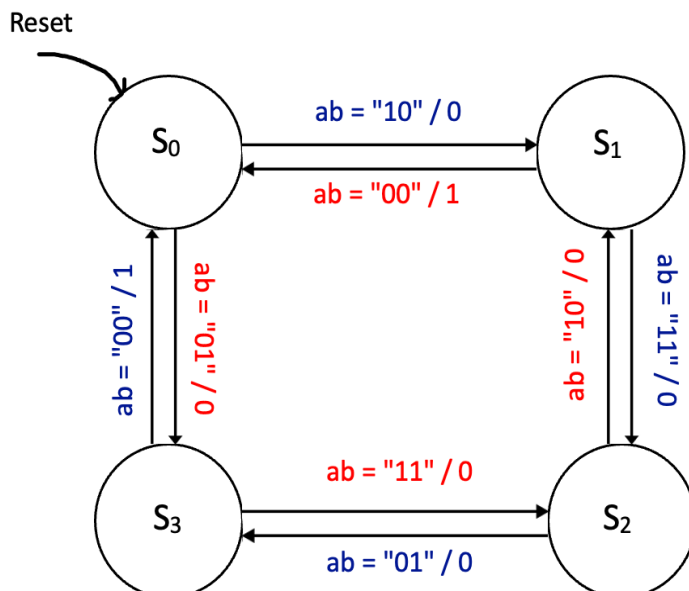
Hanie Vatani

14000122668035

برای طراحی یک سنسور ورود و خروج ماشین که تعداد ماشین های وارد شده و خارج شده را به عنوان خروجی به ما بدهد؛ می توانم از ماشین حالت برای کار استفاده کنیم. برای این برنامه نیاز به دو شمارنده مجزا به عنوان خروجی برای تعداد ماشین های وارد شده و خارج شده داریم. همچنین به ۴ ورودی اعم از Reset، CLK، sensorA و sensorB هم نیاز داریم.



ماشین حالت:



S₀: sensorA = '0', sensorB = '0' => ab = "00"

S₁: sensorA = '1', sensorB = '0' => ab = "10"

S₂: sensorA = '1', sensorB = '1' => ab = "11"

S₃: sensorA = '0', sensorB = '1' => ab = "01"

Car In Count

Car Out Count

کد برنامه :

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity exe2 is
    Port (
        Reset, CLK : std_logic;
        sensorA : in std_logic;
        sensorB : in std_logic;
        carIn : out integer range 0 to 15; --car in count
        carOut : out INTEGER range 0 to 15 --car out count
    );
end exe2;

architecture exe2 of exe2 is
    type STATE_TYPE is (S0, S1, S2, S3); --S0 is 00, S1 is 10, S2 is 11, S3 is 01

    signal ab: std_logic_vector (1 downto 0);
    signal CURRENT_STATE, NEXT_STATE: STATE_TYPE;
    signal carInCount, carOutCount: integer range 0 to 15 := 0;
begin
    CMB: process (sensorA, sensorB)
    begin
        ab <= sensorA & sensorB;
        carInCount <= carInCount;
        carOutCount <= carOutCount;
        case CURRENT_STATE is
            when S0 =>
                if (ab = "10") then --car in
                    NEXT_STATE <= S1;
                elsif (ab = "01") then --car out
                    NEXT_STATE <= S3;
                end if;
            when S1 =>
                if (ab = "11") then --car in
                    NEXT_STATE <= S2;
                elsif (ab = "00") then --car out
                    carOutCount <= carOutCount + 1;
                    NEXT_STATE <= S0;
                end if;
            when S2 =>
                if (ab = "10") then --car in
                    NEXT_STATE <= S1;
                elsif (ab = "01") then --car out
                    NEXT_STATE <= S3;
                end if;
            when S3 =>
                if (ab = "11") then --car in
                    NEXT_STATE <= S2;
                elsif (ab = "00") then --car out
                    carOutCount <= carOutCount + 1;
                    NEXT_STATE <= S0;
                end if;
            end case;
        end process;
    end;
```

```

when S2 =>
    if (ab = "01") then --car in
        NEXT_STATE <= S3;
    elsif (ab = "10") then --car out
        NEXT_STATE <= S1;
    end if;
when S3 =>
    if (ab = "00") then --car in
        carInCount <= carInCount + 1;
        NEXT_STATE <= S0;
    elsif (ab = "11") then --car out
        NEXT_STATE <= S2;
    end if;
end case;
end process CMB;

process(Reset, CLK)
begin
    if (Reset = '1') then
        CURRENT_STATE <= S0;
    elsif (CLK'event and CLK = '1') then
        CURRENT_STATE <= NEXT_STATE;
    end if;
end process;

carIn <= carInCount;
carOut <= carOutCount;

end exe2;

```

سیس کد را compile کرده و فایل test bench برای شبیه سازی می سازیم. و به آن کد زیر را اضافه می کنیم:

```

Reset <= '1', '0' after 4ns;
CLK <= '0', '1' after 2.5ns, '0' after 5ns, '1' after 7.5ns, '0' after 10ns, '1' after 12.5ns, '0' after 15ns, '1' after 17.5ns,
'0' after 20ns, '1' after 22.5ns, '0' after 25ns, '1' after 27.5ns, '0' after 30ns, '1' after 32.5ns, '0' after 35ns, '1' after 37.5ns,
'0' after 40ns, '1' after 42.5ns, '0' after 45ns, '1' after 47.5ns, '0' after 50ns, '1' after 52.5ns, '0' after 55ns, '1' after 57.5ns,
'0' after 60ns, '1' after 62.5ns, '0' after 65ns, '1' after 67.5ns, '0' after 70ns, '1' after 72.5ns, '0' after 75ns, '1' after 77.5ns,
'0' after 80ns, '1' after 82.5ns, '0' after 85ns, '1' after 87.5ns, '0' after 90ns, '1' after 92.5ns, '0' after 95ns, '1' after 97.5ns,
'0' after 100ns, '1' after 102.5ns, '0' after 105ns, '1' after 107.5ns, '0' after 110ns, '1' after 112.5ns,
'0' after 115ns, '1' after 117.5ns, '0' after 120ns, '1' after 122.5ns, '0' after 125ns, '1' after 127.5ns,
'0' after 130ns, '1' after 132.5ns, '0' after 135ns, '1' after 137.5ns, '0' after 140ns, '1' after 142.5ns,
'0' after 145ns, '1' after 147.5ns, '0' after 150ns, '1' after 152.5ns, '0' after 155ns, '1' after 157.5ns,
'0' after 160ns, '1' after 162.5ns, '0' after 165ns, '1' after 167.5ns, '0' after 170ns, '1' after 172.5ns,
'0' after 175ns, '1' after 177.5ns, '0' after 180ns, '1' after 182.5ns, '0' after 185ns, '1' after 187.5ns,
'0' after 190ns, '1' after 192.5ns, '0' after 195ns, '1' after 197.5ns, '0' after 200ns, '1' after 202.5ns,
'0' after 205ns, '1' after 207.5ns, '0' after 210ns, '1' after 212.5ns, '0' after 215ns, '1' after 217.5ns,
'0' after 220ns, '1' after 222.5ns, '0' after 225ns, '1' after 227.5ns, '0' after 230ns;

sensorA <= '0' after 10ns, '1' after 20ns, '1' after 30ns, '0' after 40ns, '0' after 50ns, '0' after 60ns,
'1' after 70ns, '1' after 80ns, '0' after 90ns, '1' after 100ns, '1' after 110ns, '1' after 120ns, '1' after 130ns,
'0' after 140ns, '0' after 150ns, '0' after 160ns, '1' after 170ns, '0' after 180ns, '1' after 190ns, '1' after 200ns,
'0' after 210ns, '1' after 220ns;

sensorB <= '0' after 10ns, '0' after 20ns, '1' after 30ns, '1' after 40ns, '0' after 50ns, '1' after 60ns,
'1' after 70ns, '0' after 80ns, '0' after 90ns, '0' after 100ns, '1' after 110ns, '0' after 120ns, '1' after 130ns,
'1' after 140ns, '0' after 150ns, '1' after 160ns, '1' after 170ns, '1' after 180ns, '1' after 190ns, '0' after 200ns,
'0' after 210ns, '0' after 220ns;

```

بعد از compile کردن فایل test bench آن را شبیه سازی می کنیم. برای این کار فایل test bench را excute می کنیم. و سپس برنامه را run می کنیم. در شبیه سازی این برنامه دنده عقب نصفه و بعد ادامه مسیر در نظر گرفته شده.

نتیجه ی شبیه سازی:

